# Eric Han

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## Education

**Stony Brook University** — Long Island, NY
*Ph.D., Applied Mathematics, Computational Track, GPA 3.81* — *2025 – Present (Expected 2030)*

**Boston University** — Boston, MA
*B.A., Applied Mathematics, GPA: 3.87* — *Sep 2023 – May 2025*

**CUNY Baruch College** — New York, NY
*B.B.A., Business Statistics, GPA: 3.97* — *Sep 2021 – May 2023, Transferred*

## Research Experience

**Air Force Research Lab** — Rome, NY
*Graduate Research Fellow; Advisor: Marco Gamarra* — *September 2026 – Present*

- Developing novel curvature-based fingerprinting methods on graphs to guide optimal network design.

*Visiting Faculty Researcher; Advisors: Marco Gamarra, Ji Liu* — *June 2026 – August 2026*

- Refined an RL pipeline maximizing algebraic connectivity $\lambda_2$ for decentralized swarm consensus via a strictly convex reward surrogate on total effective resistance $R_G$, circumventing Cauchy interlacing plateaus.
- Designed an orthogonal-invariant edge candidacy algorithm using several curvature based techniques and fast matrix updates, guaranteeing linearly independent updates, fast update computation, and strictly positive eigenvalue growth.
- Eliminated $\mathcal{O}(n^3)$ spectral recalculations via exact closed form updates for structural measures, accelerating candidate scoring by $290\times$ at machine precision.
- Developed tighter bracketing for eigenvalue search by rootfinding via Weyl's inequality.
- Benchmarked the environment against baseline models across network densities up to $n = 64$, showing consistent gains in training throughput and resulting network resilience.

**Boston University** — Boston, MA
*Undergraduate Researcher; Advisors: Caitlin Lienkaemper, Gabriel Ocker* — *June 2024 – September 2025*

- Investigated emergent dynamics in Combinatorial Threshold-Linear Networks (CTLNs), characterizing convergence to distinct dynamical regimes – from stable fixed points to chaotic attractors – in partially symmetric random graphs.
- Derived a closed-form phase transition boundary in the large network limit, analytically linking target-free clique formation probability to global fixed-point stability.
- Built a multithreaded Julia ODE framework with adaptive stiff/non-stiff solvers, vectorizing integration to scale capacity tenfold ($N = 100 \to 1{,}000$) across 1000+-trial stochastic parameter sweeps.
- Manuscript in preparation for submission to Physical Review.

**CUNY Baruch College** — New York, NY
*Undergraduate Researcher, Polymath Jr.; Advisors: Matthew Junge, Si Tang* — *May 2023 – Jan 2024*

- Analyzed the self-similar frog model on infinite $d$-ary trees to establish rigorous upper bounds on the critical drift $p_d$ required for particle recurrence.
- Bypassed combinatorial explosion by mapping vertex activations to a theoretical "pulse process," then verified exact recurrence thresholds via Sturm's theorem on the resulting expectation polynomials, proving bounds up to $d = 13$ without floating-point failure.

- Co-authored publication, "Improved critical drift estimates for the frog model on trees." *E.C.P.*, 2024. (Link)

## Technical Projects

**Physics Informed Neural Networks for Inverse PDEs** — *Nov 2025 – Dec 2025*

- Implemented Physics-Informed Neural Networks (PINNs) in PyTorch to solve nonlinear PDEs.
- Analyzed accuracy and convergence of learned solutions, identifying sensitivity to architecture and optimization dynamics.

**Numerical Methods & Solvers** — *Jan 2026 – May 2026*

- Implemented and benchmarked core numerical linear algebra algorithms and multi-dimensional finite difference simulators in C/Python to rigorously evaluate algorithmic complexity and machine precision limits.
- Extended conservative, entropy-preserving schemes using higher-order discretizations and TVD-property methods to suppress shockwave oscillations, validating numerical stability against standard LAPACK libraries.

## Publications

- Mathews Jr., P., Bailey, E., Bills, Y., Cheng, F., Chinthila, O., Dugar, A., Eser, A., **Han, E.**, Kar, C., Junge, M., Le, V., Liu, J., McDonald, Z., Tang, S., Thomas, M., Wynn, S., and Yu, E. (2024). "Improved Critical Drift Estimates for the Frog Model on Trees." *Electronic Communications in Probability.*

## Presentations

**Joint Mathematics Meetings** San Francisco, CA
*Poster: Monotonicity for the Frog Model with Drift on Trees* *January 2024*
- Authors: Bills, Y; Cheng, F; Han, E; Le, V; Wynn, S; Yu, E

**Joint Mathematics Meetings** San Francisco, CA
*Lecture: Improved Critical Drift Estimates for the Frog Model on Trees* *January 2024*
- Authors: Bills, Y; Cheng, F; Han, E; Yu, E

## Honors and Awards

**Graduate Research Fellow**
*Air Force Research Laboratory* *2026*

**Ph.D. Funding and Tuition Coverage**
*Stony Brook University* *2025 – 2027*

**Baruch Scholars**
*CUNY Baruch College* *2021 – 2023*

**Zicklin Undergraduate Honors Program**
*CUNY Baruch College* *2021 – 2023*

## Professional Experience

**Stony Brook University** Long Island, NY
*Teaching Assistant* *Aug 2025 – Present*
- Mentored 200+ students through targeted weekly office hours, translating abstract theory into structured problem-solving approaches, improving student exam performance.

## Technical Skills

**Languages**: Python, Julia, MATLAB, Mathematica, SageMath
**Tools**: LaTeX, PyTorch, LAPACK
**Core Skills**: Physics-Based Modeling, Network Optimization, Algorithm Design
**Coursework**: Real Analysis, Complex Analysis, Numerical Analysis, Numerical Linear Algebra, Finite Difference Methods, Functional Analysis, Consistency and Stability Analysis, Partial Differential Equations, Nonlinear Dynamical Systems
**Undergraduate Coursework**: Methods of Applied Math 1, Methods of Applied Math 2, Modern Algebra 1, Probability, Real Analysis